TuteDude

WhoAmI?

- Senior Information Security Analyst (Team Lead) at a CERT-In Empanelled Organization
- eCIR | CHFI | ISC2 CC | An Alumni of NFSU
- 3+ years in SOC Operations, Incident Response, Cloud Security, and VAPT
- Trained 500+ students & professionals in Cybersecurity and Forensics
- Developed CTF challenges and cybersecurity labs for advanced training

PRESENTATION BY HARSHIT BHARDWAJ

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Digital Era

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2016 US Elections & Cambridge Analytica

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the unethical data collection.

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What data was collected?

- Where users lived
- What pages users liked
- Psychographic profiles of users
- Private messages
- Characteristics and personality traits like openness, conscientiousness, agreeableness, IQ, gender, age, political views, and religion *(Hartmans)*.

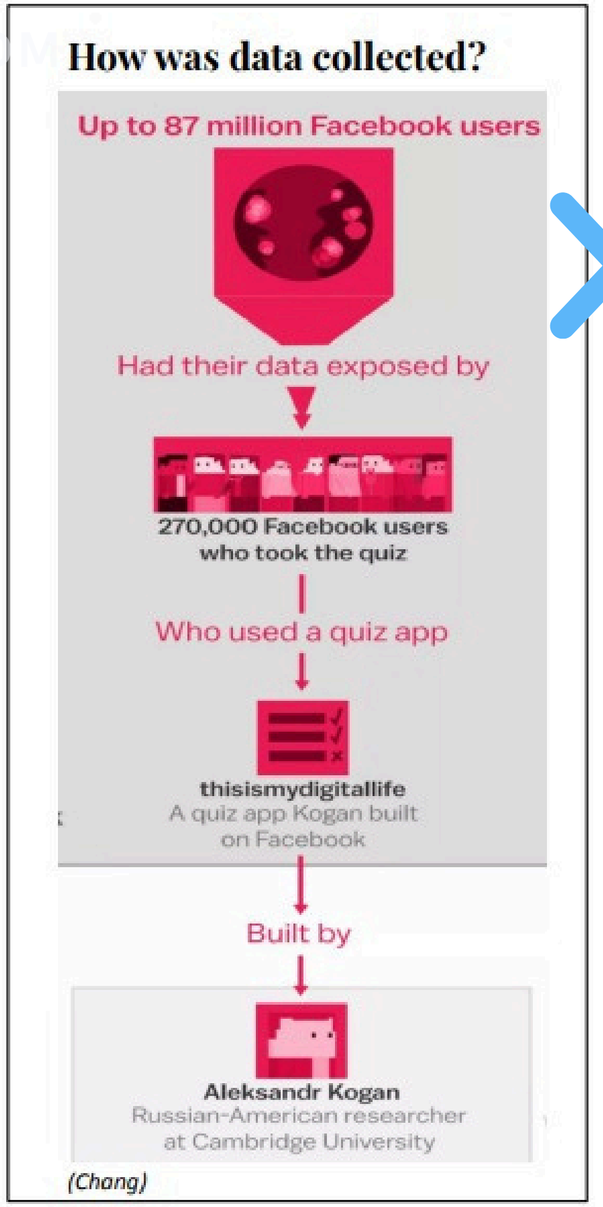
How was the data processed?

The quiz app had survey questions which were turned into insights for voting purposes, using the 'oceans model' harvesting more than 5,000 data points on each profile and categorizing them accordingly based on their answers and their attitude towards other pages and posts *(Chang)*.

Where it went wrong?

Point of view #1

The loophole in the quiz app "thisismydigitallife" which allowed access to the profiles of participants friends is where it all went wrong, we need to understand that the consent was present for these initial participants of the quiz app, but not that of their friends.



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Cyber Attacks

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What is CyberSecurity?

CyberSecurity is the body of technologies, processes and practices designed to protect networks, computers, programs and data from attack, damage or unauthorized access

- Application Security
- Information Security
- Network Security
- Operational Security
- Disaster Security
- End-user Education

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CIA Triad

Confidentiality

Confidentiality is about preventing the disclosure of data to unauthorized parties.

Integrity

Integrity refers to protecting information from being modified by unauthorized parties.

Availability

Availability is making sure that authorized parties are able to access the information when needed

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Three Pillars of CyberSecurity

People

Processes

Technology

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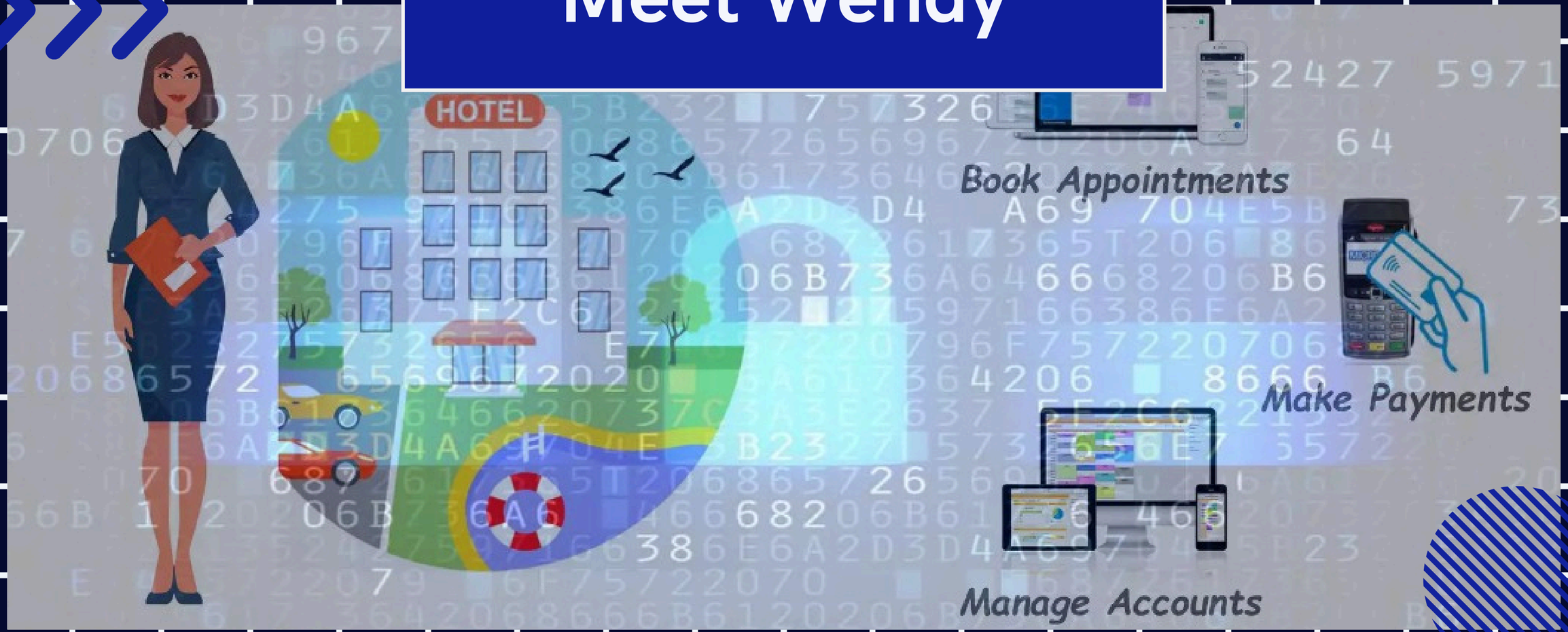
Meet Wendy



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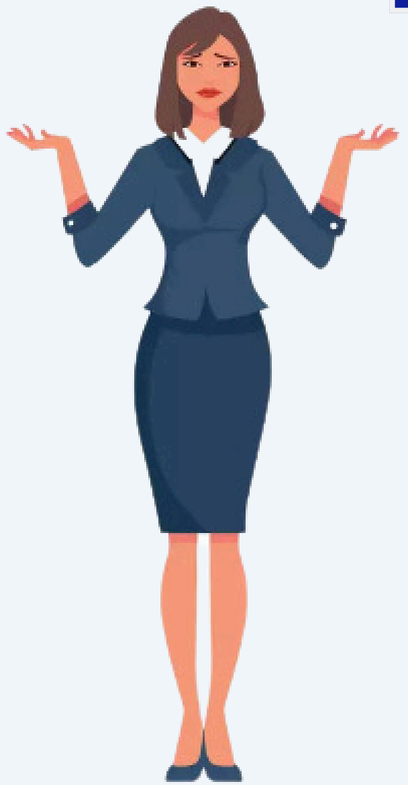
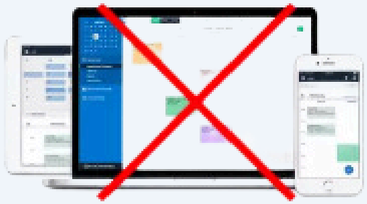
Incident





Book Appointments

Make Payments

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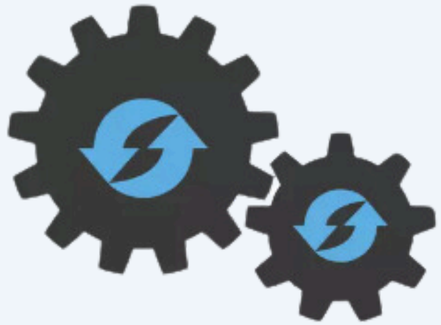
Incident Response Team



Active Response System

- Identify the source and performs business impact analysis
- Detects logs and access control issues
- Contains the threat paths to stop further damage
- Prioritizes recovery and install prevention software

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What You Will Learn

- Python and Linux Fundamentals
- Foundations of Information Security
- Application Security and Penetration Testing
- Network Defense & Penetration Testing
- Governance, Risk & Compliance (ISO, SMS)
- Securing Emerging Technologies (Blockchain, AI, IoT, Cloud)
- Endpoint Hardening and Incident Handling Concepts
- Security Operations Center: Overview & Roles
- SIEM Architecture & Hands-On Splunk
- RCA (Root Cause Analysis) & Cyber Breach Investigation

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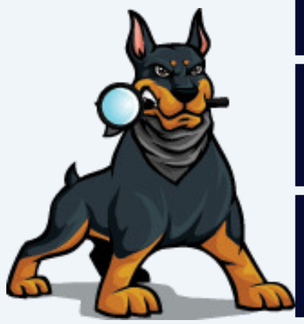
Tools We'll be Working On

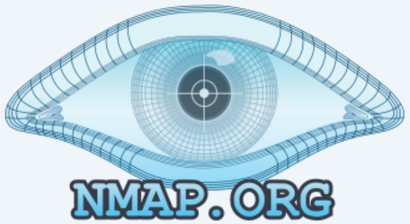
ANY RUN
INTERACTIVE MALWARE ANALYSIS

splunk>

BURPSUITE

yara





Nessus®
vulnerability scanner

CYBER TRIAGE

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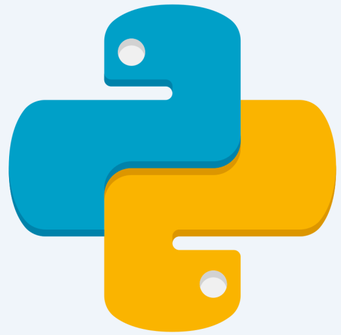
Python & Linux Fundamentals

Python for Cyber Security

- Basics of Python Programming: variables, data structures, functions
- Web Scraping Tools: Scrapy, Beautiful Soup
- Network Scripting and Tools: Twisted, Nmap



Linux for Cyber Security

- Linux Command Line Basics
- File Permissions and Ownership
- Shell Scripting for Automation

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Foundations of Information Security

Understanding Cyber Threats

- Types of Threats: hacking, phishing, ransomware
- Ethical Hacking Tools

Foundations of Information Security

- CIA Triad: Confidentiality, Integrity, Availability
- Techniques for Information Gathering and Reconnaissance

Common Cyber Attacks

- Denial of Service (DoS), Distributed DoS
- Spoofing, Tampering, Data Breaches

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Application Security and Penetration Testing

Securing Web Applications

- Common Attacks: SQL Injection, Cross-Site Scripting
- Practical Exercises with Security Tools: Metasploit, BWAPP, DVWA

Mobile Application Security

- Types of Mobile Attacks and Prevention
- Network Reconnaissance and Vulnerability Assessment

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Network Defense & Penetration Testing

Fundamentals of Network Security

- Security Controls and Protocols
- Deep Packet Inspection and Intrusion Detection Systems
- Configuring Network Devices: Firewalls, Routers, Switches

Data Security Measures

- Encryption and Decryption Techniques
- Data Backup Strategies and Storage Solutions

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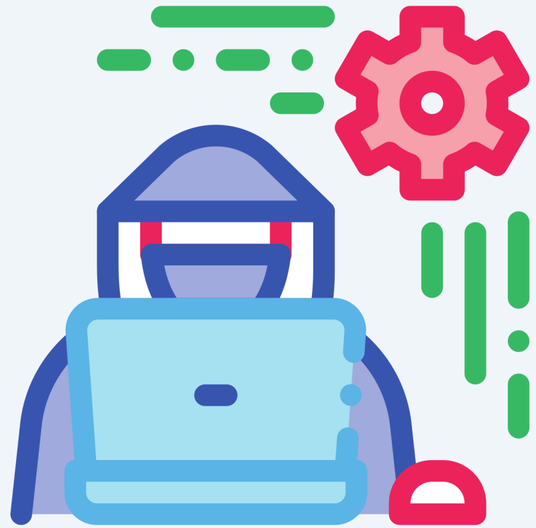
Network Defense & Penetration Testing

Conducting Vulnerability Assessments

- Types of Assessments: Host, Network, Application
- Tools and Techniques for Identifying Vulnerabilities

Penetration Testing

- Planning and Executing Pen Tests
- Server Hardening and Attack Prevention



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Governance, Risk & Compliance (ISO, ISMS)

- Security Audits and Compliance
  - Performing Security Audits and Assessments
  - Importance of Compliance and Risk Management

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What is Kali Linux ?

Kali Linux is an operating system Debian-derived Linux distribution designed for digital forensics and penetration testing (ethical hacking). It is maintained and funded by Offensive Security

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Different CyberSecurity Job Profiles

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Computer Forensic Analyst

- A computer forensics analyst, also known as a digital forensics examiner, specializes in the recovery and investigation of material found in digital devices.
- This role is crucial in legal contexts, corporate investigations, and cybersecurity incident response.

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Key Responsibilities

- **Evidence Collection**

Securing and collecting digital evidence from various devices (computers, mobile devices, storage media).

- **Data Recovery**

Retrieving data from damaged, corrupted, or compromised devices.

- **Analysis**

Examining digital data to uncover and document how a breach or crime occurred.

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Key Responsibilities

- Documentation

Maintaining detailed records of findings and processes to ensure evidence integrity.

- Reporting

Creating comprehensive reports for legal proceedings, internal investigations, or regulatory compliance.

- Legal Testimony

Providing expert testimony in court about the methods and findings of forensic analysis.

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Technical Skills

- Operating Systems: Proficiency in Windows, Linux, and macOS.
- File Systems: Understanding of various file systems (NTFS, FAT32, EXT4).
- Forensic Tools: Expertise in tools like EnCase, FTK, Autopsy, and Sleuth Kit.
- Data Recovery Techniques: Skills in recovering lost, deleted, or encrypted data.
- Networking: Basic knowledge of network protocols and analysis.
- Certifications: CHFI, Certified Fraud Examiner, GIAC Certified Fraud Analyst

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Industry Applications

Law Enforcement

- Assisting in criminal investigations and prosecutions.

Corporate Security

- Investigating internal incidents, data breaches, and policy violations.

Legal and Consulting Firms

- Providing expertise in digital investigations for legal cases and corporate clients.

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Tools and Technologies

EnCase Guidance Software

AUTOPSY DIGITAL FORENSICS

ACCESSDATA ForensicToolkit (FTK)

Cellebrite UFED

VOLATILITY

OXYGEN FORENSICS

CROWDSTRIKE

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SOC Analyst

- A SOC Analyst works within the Security Operations Center (SOC) to monitor, detect, and respond to security threats and incidents.

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Key Responsibilities

- **Monitoring Security Alerts:**

Continuously monitor security alerts from systems such as firewalls, IDS/IPS, and SIEM tools.

- **Incident Detection and Response:**

Rapid identification and response to security incidents to mitigate potential damage.

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SOC Analyst Career Path

- Progression:

L1: Entry-level, foundational monitoring and triage.

- Monitor security alerts.
- Perform initial triage to determine alert validity.

L2: Advanced analysis and incident response.

- Conduct detailed incident investigations.
- Coordinate incident response efforts.

L3: Leadership in threat intelligence and incident management.

- Lead major incident response efforts.

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SOC Analyst Tools

- **Security Information and Event Management (SIEM):**

Tools like Splunk, IBM QRadar for centralized logging and real-time analysis.

- **Intrusion Detection Systems (IDS)/Intrusion Prevention Systems (IPS):**

Tools like Snort are used for identifying and preventing potential threats.

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SOC Analyst

- **Firewalls and Antivirus Software:**

Tools for network defense and malware protection.

- **Endpoint Detection and Response (EDR):**

Tools like CrowdStrike for endpoint monitoring and threat detection.

CROWDSTRIKE

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Technical Skills

- **Networking:**

Deep understanding of network protocols, architecture, and operations.

- **Operating Systems:**

Proficiency in Windows, Linux, and macOS security measures.

- **SIEM Proficiency:**

Expertise in configuring and using SIEM tools to analyze security events.

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Technical Skills

- **Threat Intelligence:**

Knowledge of threat landscapes, common attack vectors, and adversarial tactics.

- **Scripting and Automation:**

Skills in scripting languages (Python, PowerShell) to automate repetitive tasks and improve efficiency.

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GRC Analyst

- A GRC Analyst focuses on ensuring that an organization's policies and procedures align with regulations, managing risks, and maintaining compliance with industry standards and legal requirements.

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Key Responsibilities

- **Policy Development**

Drafting and updating policies to ensure compliance with regulations.

- **Risk Assessments**

Conducting regular risk assessments and implementing mitigation strategies.

- **Compliance Audits**

Performing internal audits and coordinating external audits.

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Essential Skills

- Essential Skills

Knowledge of Regulations and Standards

Familiarity with GDPR, HIPAA, PCI-DSS, etc.

- Communication Skills

Excellent written and verbal communication for reporting and policy development.

- Technical Proficiency

Understanding of IT systems, security controls, and data privacy.

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Career Path and Development

- **Entry-Level Positions**

Risk Analyst, Compliance Analyst, IT Auditor

- **Mid-Level Positions**

Senior GRC Analyst, GRC Manager

- **Advanced Positions**

Chief Risk Officer (CRO), Chief Compliance Officer (CCO)

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Penetration Tester (Ethical Hacker)

A Penetration Tester, often referred to as an Ethical Hacker, is a cybersecurity professional who simulates cyberattacks on an organization's systems, networks, and applications.

The goal is to identify and exploit vulnerabilities before malicious hackers can discover and exploit them.

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Penetration Tester (Ethical Hacker)

- Planning and Scoping Tests: Collaborating with stakeholders to understand testing requirements, define the scope of the test, and identify target systems.

- Conducting Penetration Tests: Using various tools and techniques to probe for vulnerabilities in web applications, networks, and other IT infrastructure.

- Exploitation: Attempting to exploit identified vulnerabilities to determine their potential impact.

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Nessus vulnerability scanner

NMAP.ORG

Essential Skills

- Technical Knowledge: Deep understanding of operating systems, network protocols, and web applications.
- Tool Proficiency: Expertise in using penetration testing tools like Metasploit, Nmap, Burp Suite, Wireshark, and OWASP ZAP.
- Reporting Findings: Documenting vulnerabilities, providing detailed analysis, and offering recommendations for remediation.

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Penetration Tester (Ethical Hacker)

- **Follow-up Testing: Retesting after fixes are applied to ensure vulnerabilities have been effectively addressed.**

- **Certifications: Professional certifications such as Certified Ethical Hacker (CEH), Offensive Security Certified Professional (OSCP), and GIAC Penetration Tester (GPEN) are highly regarded.**

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Other CyberSecurity Job Roles Include

- Incident Responder: Handles and mitigates cyber incidents by investigating breaches and coordinating response efforts.
- Security Architect: Designs secure network systems and develops security policies and frameworks.
- Malware Analyst: Studies and dissects malicious software to understand its behavior, origin, and impact, and to develop methods for detection and removal.

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Red Team Vs Blue Team

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What is Red Teaming?

Red Team security isthename forcybersecurity professionals who act as adversaries against malicious hackers. Red Teams often consist of independent ethical hackers who objectively evaluate system security

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Why do you need Red Team Professionals

- These experts identify breaches through real-world attack techniques in a simulated environment.
- The Red Team relies on the adversarial approach because the defense of IT infrastructure cannot rely on a security systems' theoretical capabilities.
- Security is only possible in the presence of real-world threats.

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What is Blue Team Security

- The Blue Team refers to the security professionals responsible for defending an IT infrastructure against a possible attack. These professionals analyze data flow and user behavior to predict a threat and take proper security measures.

- They are aware of the business objectives and the organization's security strategy. Therefore, their task is to strengthen the castle walls so no intruder can compromise the defenses

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What is Blue Team Security

- Blue teams must establish security measures around the key assets of an organization. They start their defensive plan by identifying the critical assets, documenting the importance of these assets to the business, and the impact the absence of these assets will have.

- Blue Team security professionals prevent cyberattacks by regularly performing checks on the system and analyzing specific internal and external network patterns.

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WannaCry Ransomware Attack Simulation

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Thank you

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